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Kusano et al.

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(54) **LIQUID EJECTION HEAD AND INKJET RECORDING APPARATUS**

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B41J 2/165 (2006.01)

(52) **U.S. Cl.**
CPC ***B41J 2/1433*** (2013.01); ***B41J 2/16538*** (2013.01); *B41J 2/14072* (2013.01); *B41J 2002/14411* (2013.01); *B41J 2002/14443* (2013.01)

(58) **Field of Classification Search**
CPC .. B41J 2/1433; B41J 2/16538; B41J 2/14072; B41J 2002/14411; B41J 2002/14443
See application file for complete search history.

(56) **References Cited**

U.S. PATENT DOCUMENTS

6,834,937 B2 * 12/2004 Killmeier B41J 2/14072 347/50
2023/0121567 A1 4/2023 Chida et al.
2024/0286408 A1 8/2024 Chida et al.

FOREIGN PATENT DOCUMENTS

JP 07-17045 A 1/1995
JP 10-44420 A 2/1998

* cited by examiner

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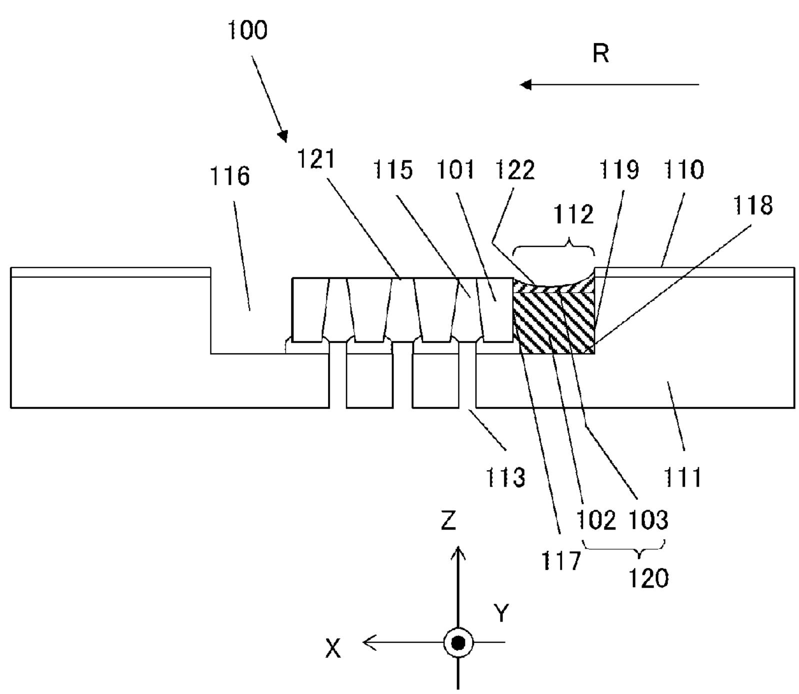
(57) **ABSTRACT**

A liquid ejection head includes: a channel member; an ejection substrate being disposed in a recess of the channel member and having a nozzle row of nozzles arranged in a first direction; and a sealing member having a thermosetting resin filled in a predetermined space defined by side and bottom surfaces of the recess and a side surface of the ejection substrate and located upstream of the ejection substrate in a second direction. The sealing member includes: a first sealing portion having a first resin located near the bottom surface of the predetermined space; and a second sealing portion having a second resin located near an ejection surface of the ejection substrate. A coefficient of linear expansion of the first resin is less than that of the second resin, and a specific gravity of the first resin is greater than that of the second resin.

13 Claims, 7 Drawing Sheets



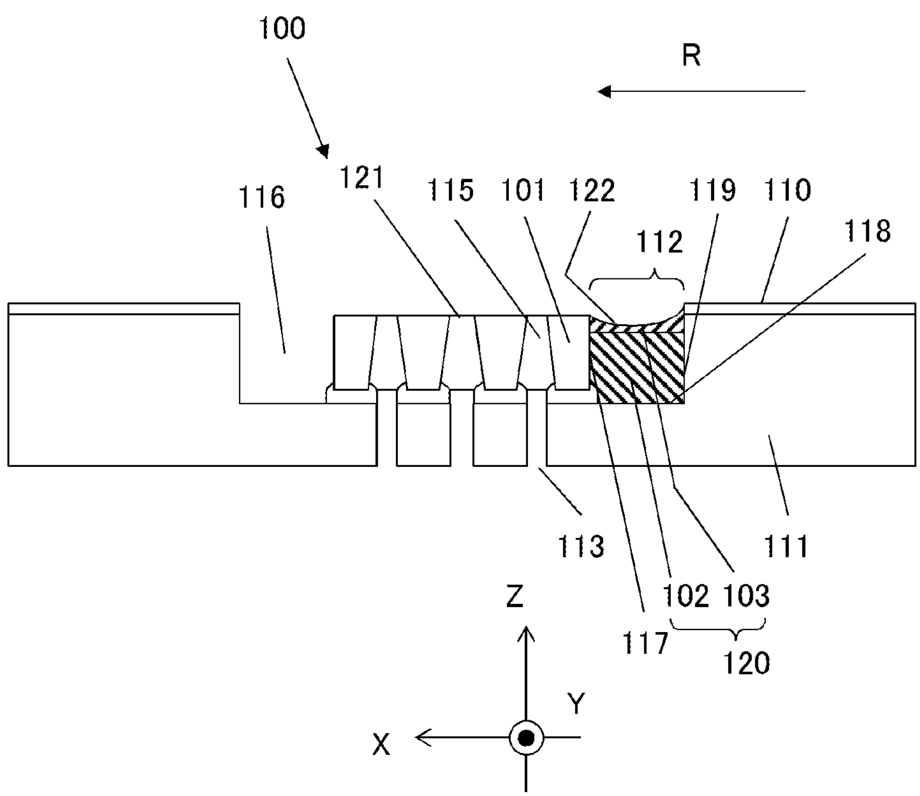

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Fig.1

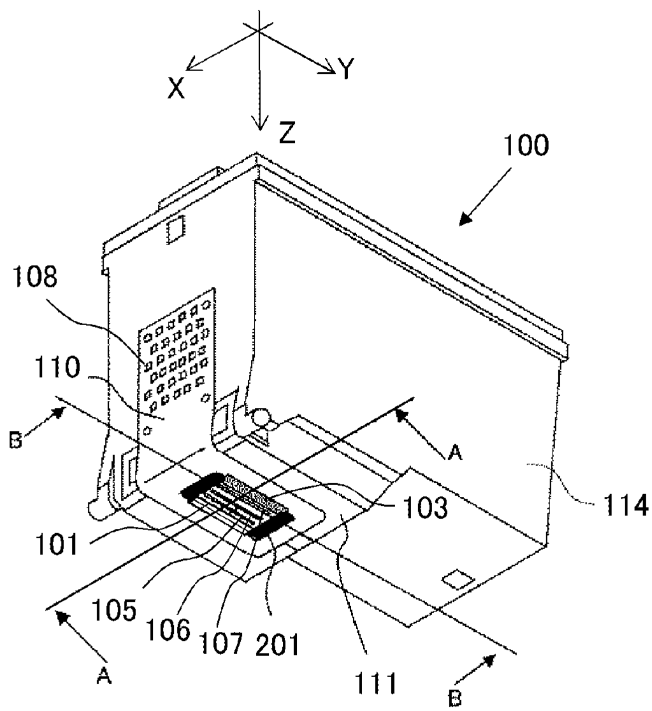


Fig.2A

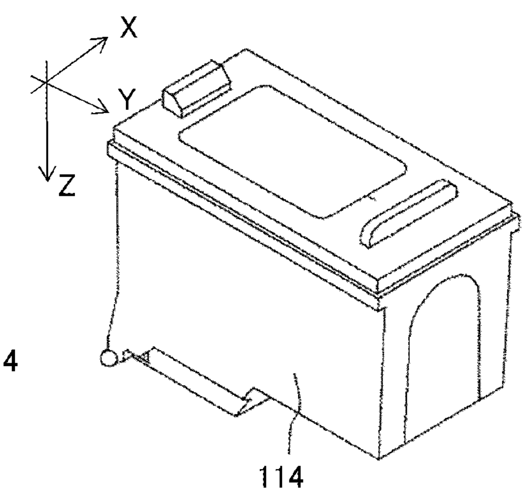


Fig.2B

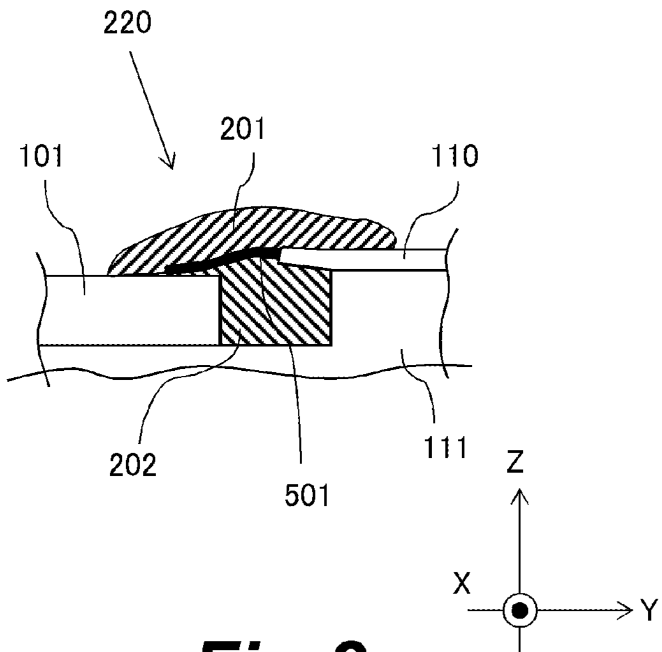


Fig.3

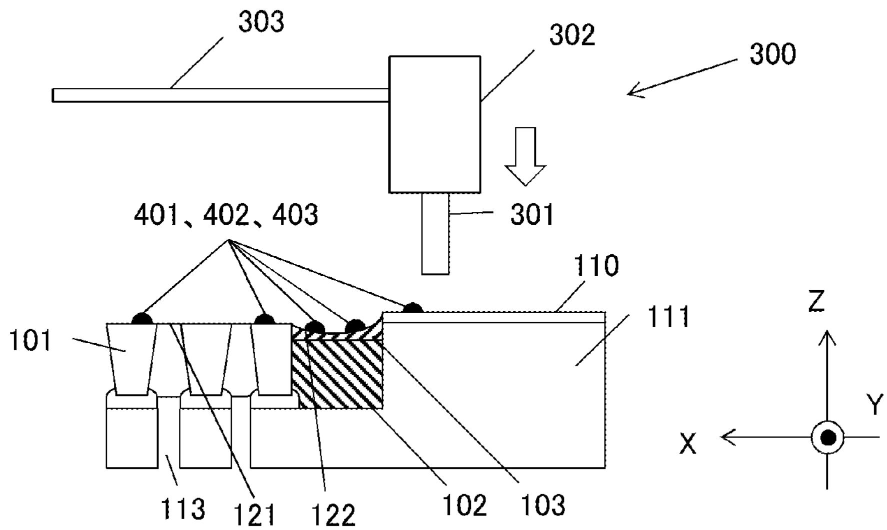


Fig.4A

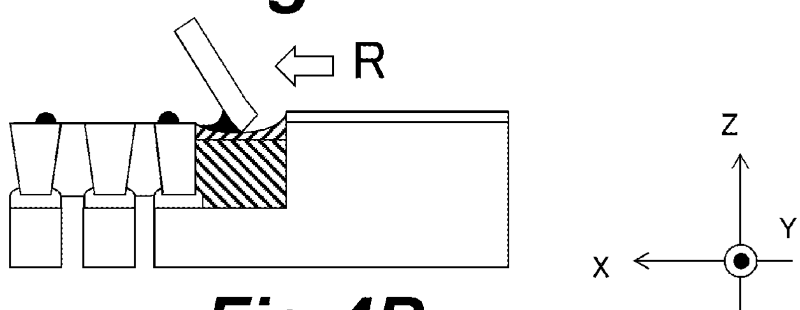


Fig.4B

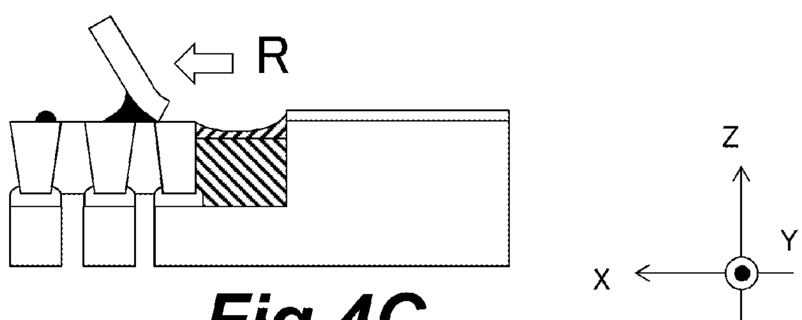


Fig.4C

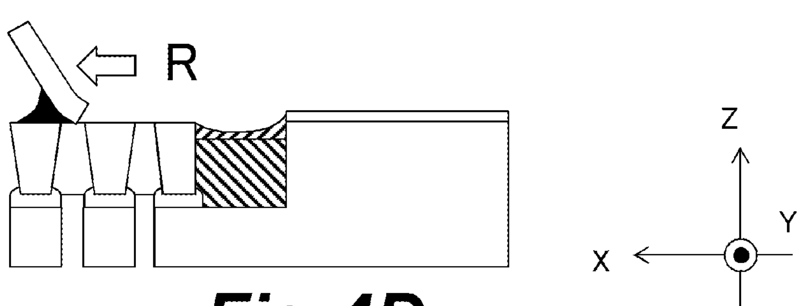
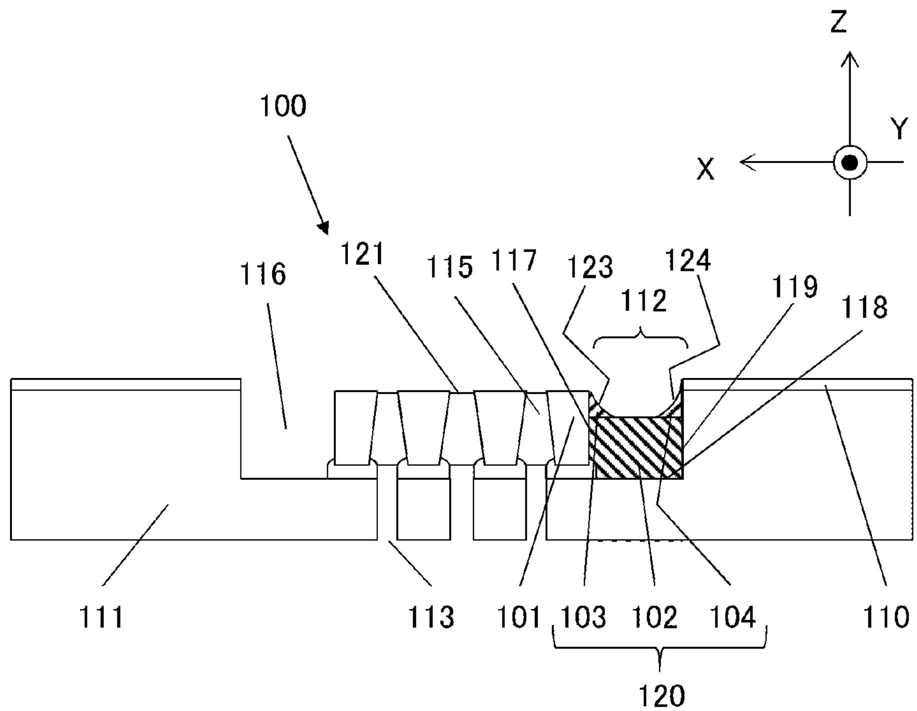


Fig.4D


Z
Y
X
100
116
121
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103
102
104
110
120

Fig.5A

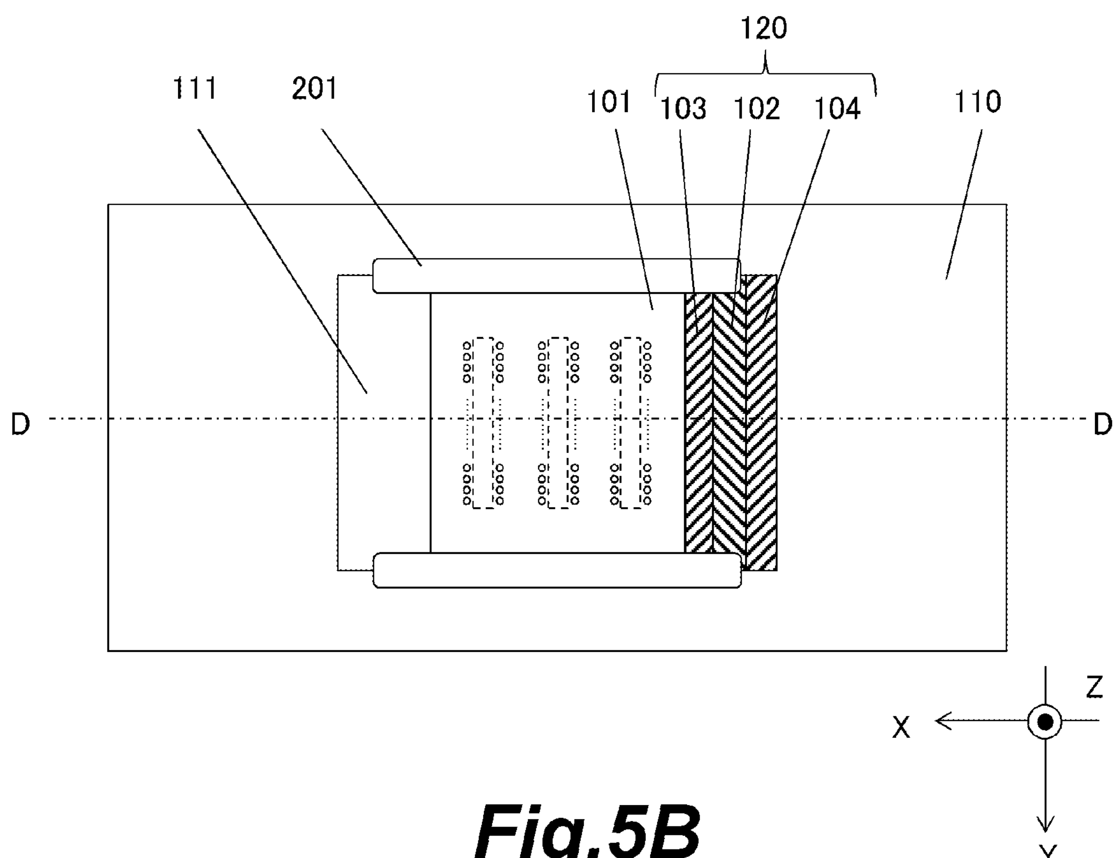
120
111
201
101
103
102
104
110
D
D
Z
X
Y

Fig.5B

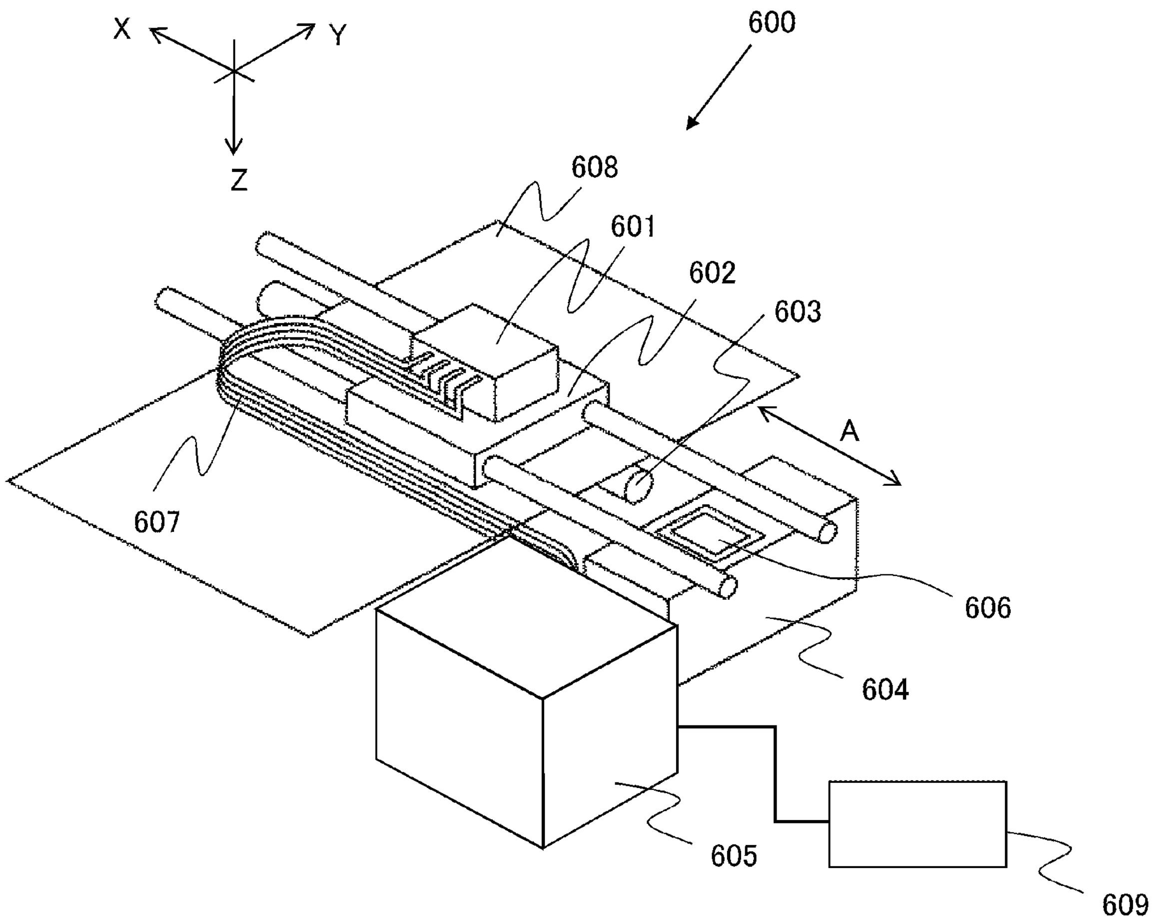

Fig.6

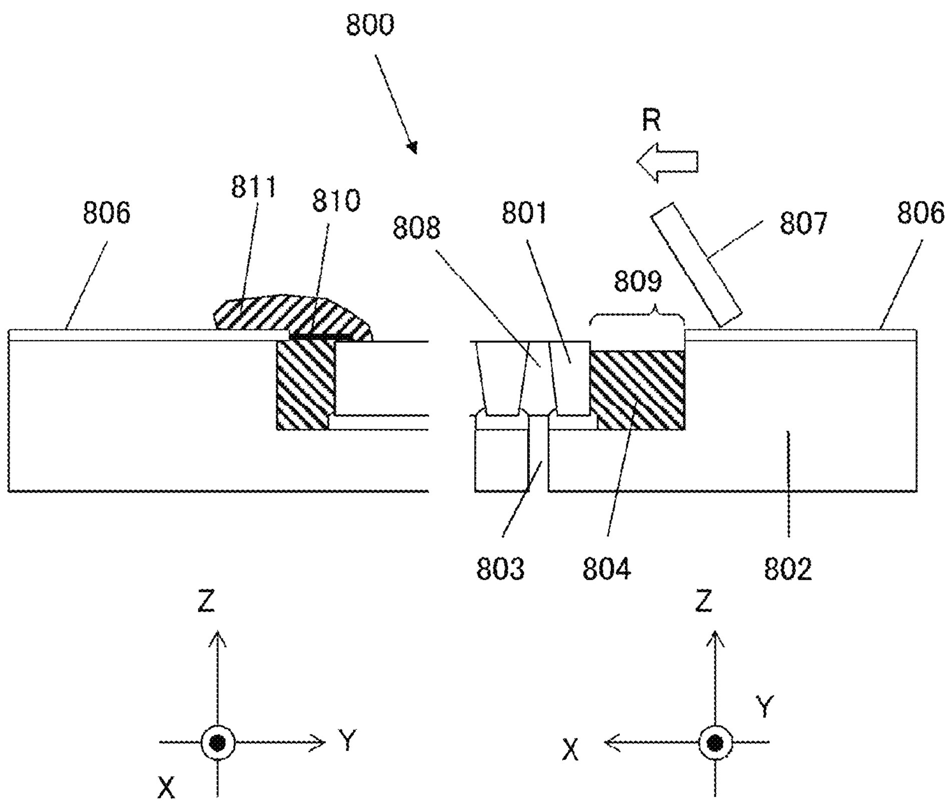

Fig.7A

Fig.7B

PRIOR ART

PRIOR ART

LIQUID EJECTION HEAD AND INKJET RECORDING APPARATUS

BACKGROUND OF THE INVENTION

Field of the Invention

The present invention relates to a liquid ejection head and an inkjet recording apparatus.

Description of the Related Art

An inkjet recording apparatus uses a liquid ejection head including nozzle rows each having a plurality of nozzles. FIG. **7**B is a schematic cross-sectional view of a liquid ejection head **800** showing a cross section perpendicular to the nozzle arrangement direction in a nozzle row (Y direction). FIG. **7**A is a schematic cross-sectional view of the liquid ejection head **800** showing a cross section perpendicular to a direction (X direction) intersecting with the nozzle arrangement direction. A channel member **802** includes a liquid channel **803** communicating with nozzles **808** of the ejection substrate **801**. Ink supplied to the liquid channel **803** from an ink tank (not shown) is ejected from the nozzle **808** in the Z direction. An electric wiring member **806**, which includes an external wiring for inputting an electric signal from the outside to the ejection substrate **801**, is fixed to the surface of the channel member **802**. The electric wiring member **806** is connected to the ejection substrate **801** by an electrical connection portion **810**. The ejection substrate **801** is disposed in a recess disposed in the channel member **802**, and a space **809** surrounded by the side surface and the bottom surface of the recess disposed in the channel member **802** and the side surface of the ejection substrate **801** is filled with a sealing member **804**. A sealing member **811** for protecting the electrical connection portion **810** is also provided at the location where the electrical connection portion **810** is provided. Japanese Patent Application Publication No. H10-44420 describes a configuration of a liquid ejection head of an inkjet recording apparatus.

In an inkjet recording apparatus including such a liquid ejection head, paper dust generated from the recording paper during the printing process, dust floating in the air, splashes of ejected ink droplets, and the like may adhere to the ejection surface including the nozzles **808** of the ejection substrate **801**. In order to remove these adherents, the inkjet recording apparatus includes a blade **807** to wipe the ejection surface. The blade **807** is made of an elastic material, such as rubber, and movable relative to the ejection surface of the ejection substrate **801**. The moving direction R of the blade **807** may be a direction along the nozzle arrangement direction (Y direction) or a direction intersecting with the nozzle arrangement direction (X direction). The wiping is described in Japanese Patent Application Publication No. H07-17045.

SUMMARY OF THE INVENTION

Since the sealing member **804** is formed by filling the space **809** with a thermosetting resin and curing it at a high temperature, the cured resin shrinks when it returns to normal temperature. Also, the resin may deform to further shrink depending on the use environment, such as when the inkjet recording apparatus is placed in a low-temperature environment. Although the ejection substrate **801** also shrinks in such an environment, the difference in coefficient of linear expansion between the ejection substrate **801** and the sealing member **804** generates tensile stress in the ejection substrate **801**. This stress may cause the ejection substrate **801** to crack.

As countermeasures against the above, the tensile stress acting on the ejection substrate **801** may be reduced by providing the sealing members **804** and **811** at the portion that protects the electrical connection portion **810** so that the sealing member **804** is not provided in the space **809**.

In this case, foreign matter such as paper dust, dust, and splashes of ink droplets may accumulate in the space **809**. This may cause foreign matter to adhere to the tip of the blade **807** when the tip of the blade **807** enters the space **809** during wiping operation. In this case, when the blade **807** passes over the ejection surface of the ejection substrate **801**, foreign matter may adhere to and remain on the ejection surface, affecting the ink ejection performance.

It is an object of the present invention to limit cracking of the ejection substrate of a liquid ejection head having a configuration in which the ejection substrate is disposed in a recess of a channel member and a space between the channel member and the ejection substrate is filled with a sealing member.

According to an aspect of the present invention, a liquid ejection head includes:

a channel member including a liquid channel;

an ejection substrate disposed on a bottom surface of a recess disposed in the channel member, the ejection substrate including a nozzle row having a plurality of nozzles communicating with the liquid channel, the nozzles being arranged in a first direction; and

a sealing member having a thermosetting resin filled in a predetermined space within a space defined by a side surface of the recess, the bottom surface of the recess, and a side surface of the ejection substrate, and the predetermined space being located upstream of the ejection substrate in a second direction intersecting with the first direction, wherein

the sealing member includes:

a first sealing portion having a first resin located near the bottom surface within the predetermined space; and

a second sealing portion having a second resin located near an ejection surface of the ejection substrate within the predetermined space, the ejection surface being a surface provided with the nozzles,

a coefficient of linear expansion of the first resin is less than a coefficient of linear expansion of the second resin, and

a specific gravity of the first resin is greater than a specific gravity of the second resin.

According to another aspect of the present invention, a liquid ejection head includes:

a channel member including a liquid channel;

an ejection substrate disposed on a bottom surface of a recess disposed in the channel member, the ejection substrate including a nozzle row having a plurality of nozzles communicating with the liquid channel, the nozzles being arranged in a first direction; and

a sealing member having a thermosetting resin filled in a predetermined space within a space defined by a side surface of the recess, the bottom surface of the recess, and a side surface of the ejection substrate, the predetermined space being located upstream of the ejection substrate in a second direction intersecting with the first direction, wherein

the sealing member includes:

a first sealing portion having a first resin located near the bottom surface within the predetermined space; and

a second sealing portion having a second resin located near an ejection surface of the ejection substrate within the predetermined space, the ejection surface being a surface provided with the nozzles, and the second sealing portion being in contact with the side surface of the ejection substrate; and

a third sealing portion having a third resin located near the ejection surface of the ejection substrate within the predetermined space, the third sealing portion being in contact with the side surface of the recess,

a coefficient of linear expansion of the first resin is less than a coefficient of linear expansion of the second resin and a coefficient of linear expansion of the third resin, and

a specific gravity of the first resin is greater than a specific gravity of the second resin and a specific gravity of the third resin.

According to the present invention, it is possible to limit cracking of an ejection substrate of a liquid ejection head having a configuration in which the ejection substrate is disposed in a recess of a channel member and a space between the channel member and the ejection substrate is filled with a sealing member.

Further features of the present invention will become apparent from the following description of exemplary embodiments with reference to the attached drawings.

BRIEF DESCRIPTION OF THE DRAWINGS

FIG. **1** is a schematic partial cross-sectional view showing the vicinity of an ejection substrate of a liquid ejection head of a first embodiment;

FIGS. **2**A and **2**B are perspective views of the liquid ejection head of the first embodiment;

FIG. **3** is a schematic partial cross-sectional view showing the vicinity of an electrical connection portion of the ejection substrate of the liquid ejection head of the first embodiment;

FIGS. **4**A to **4**D are diagrams showing a wiping operation of the ejection substrate of the liquid ejection head of the first embodiment;

FIGS. **5**A and **5**B are a schematic partial cross-sectional view and a plan view showing the vicinity of an ejection substrate of a liquid ejection head of a second embodiment;

FIG. **6** is a diagram showing the schematic configuration of an inkjet recording apparatus according to an embodiment; and

FIGS. **7**A and **7**B are schematic partial cross-sectional views showing the vicinity of an ejection substrate of a conventional liquid ejection head.

DESCRIPTION OF THE EMBODIMENTS

Referring to the drawings, embodiments of the present invention are now described.

First Embodiment

FIG. **6** is a perspective view schematically showing the configuration of an inkjet recording apparatus according to a first embodiment of the present invention. The inkjet recording apparatus **600** performs recording on a recording medium **608** by repeating reciprocating movement (main scanning) of a liquid ejection head **601** (recording head) in the X directions and transport (sub-scanning) of the recording medium **608**, such as a sheet, in a Y direction at a predetermined pitch. In synchronization with these movements, the inkjet recording apparatus **600** selectively ejects liquid (pigment ink) of different colors from the liquid ejection head **601** to cause the ink to land on the recording medium **608**, thereby forming characters, symbols, images, and the like on the recording medium **608**. The recording medium **608** may be any medium that allows ink droplets to land thereon to form images. The recording medium **608** may be of various materials and forms, such as paper, fabric, optical disc label surface, plastic sheet, OHP sheet, and envelope.

Referring to FIG. **6**, the liquid ejection head **601** is mounted on a carriage **602** in a detachable manner. The carriage **602** is slidably supported by two guide rails extending in the X direction. A driving means such as a motor (not shown) linearly reciprocates the carriage **602** in the X directions along the guide rails. A transport roller **603** serving as a transport means transports the recording medium **608** in a Y direction intersecting with the moving direction of the carriage **602** (X direction), with the recording medium **608** facing the liquid ejection surface of the liquid ejection portion of the liquid ejection head **601**. The recording medium **608** thus receives the liquid ejected from the liquid ejection head **601**.

The liquid ejection head **601** includes, as a plurality of liquid ejection portions, a plurality of nozzle rows for ejecting liquids of different colors (for example, yellow, magenta, and cyan inks). Liquids of different colors are independently supplied from the liquid supply unit **605** to the respective nozzle rows of the liquid ejection head **601** through respective liquid supply tubes **607**.

In a non-recording region A, which is within the range of reciprocating movement of the liquid ejection head **601** in the X direction and outside the range in which the recording medium **608** passes, a recovery unit **604** is arranged to face the ink ejection surface of the liquid ejection head **601**. The recovery unit **604** includes a cap portion for capping the liquid ejection surface of the liquid ejection head **601**, and a suction mechanism for performing forcible suction of liquid with the liquid ejection surface capped. The recovery unit **604** also includes a wiper **606** including a cleaning blade for removing smears on the liquid ejection surface. The operation of the wiper **606** will be described below. The suction is performed by the recovery unit **604** prior to the recording operation of the inkjet recording apparatus **600**. As such, even when the inkjet recording apparatus **600** is operated after being left unused for a long period of time, the recovery process performed by the recovery unit **604** removes residual bubbles in the liquid ejection portion of the liquid ejection head **601** and thickened liquid near the nozzles. This limits changes in the ejection performance of the liquid ejection head **601**.

The inkjet recording apparatus **600** has a control portion **609**, which controls the operation of each portion of the inkjet recording apparatus **600**. The control portion **609** includes a CPU (not shown), a memory for storing programs and the like, an input/output circuit, and the like. The control portion **609** includes an input/output circuit and the like that control the reception of data, such as images to be recorded on the recording medium **608**, from an external device, the operation of the motor that drives the carriage **602** and the transport roller **603**, the liquid ejection operation by the liquid ejection head **601**, and the operation of the wiper **606**.

FIGS. **2**A and **2**B are perspective views of a liquid ejection head **100** of the first embodiment. The direction along line B-B in FIG. **2**A is parallel to the arrangement direction (Y direction) of the nozzles in a nozzle row of the liquid ejection head **100**, and the direction along line A-A in FIG. **2**A is parallel to the direction in which nozzle rows **105**, **106**, and **107** of yellow, magenta, and cyan are arranged (X direction). In the first embodiment, line A-A is perpendicular to line B-B.

The liquid ejection head **100** has a configuration in which an ejection substrate **101** and an ink container portion **114** containing ink are integrated. The ejection substrate **101** has a heater, serving as an energy generating element for ejecting ink, and a substrate (element substrate) including wiring for transmitting electrical energy supplied from the inkjet recording apparatus to the heater. On the substrate, a nozzle plate (channel forming member) is provided that includes channels for supplying ink to the heater and nozzles (ejection ports) for ejecting ink.

The nozzle plate of the ejection substrate **101** includes nozzle rows **105**, **106**, and **107** for ejecting ink of three colors, yellow, magenta, and cyan.

The ink to be supplied to the ejection substrate **101** is contained in an ink container portion **114**, which holds and stores ink of each color. Ink absorbers for holding ink and ink supply passages for supplying ink to the ejection substrate **101** are provided in the ink container portion **114**. The liquid ejection head **100** has a channel member **111** for supplying ink from the ink container portion **114** to the ejection substrate **101**. Ink is supplied from the ink supply passage to the nozzle rows of the ejection substrate **101** through the channel member **111**. The ink supply passage has a filter for limiting entry of foreign matter into the nozzles.

The ejection substrate **101** is made of a silicon substrate and is bonded and fixed to the bottom surface of a recess disposed in the channel member **111**. Within the space (gap) surrounded by the side surface of the ejection substrate **101** and the side surface and the bottom surface of the recess of the channel member **111**, the space on the upstream side in the moving direction of the blade during wiping is sealed by a first sealing portion **102** and a second sealing portion **103**. Note that the space on the downstream side in the blade moving direction within the above space may also be sealed by the first sealing portion **102** and the second sealing portion **103**. As will be described below, in the configuration of the first embodiment, the second sealing portion **103** is placed over the first sealing portion **102** in the Z direction (ejection direction), and the second sealing portion **103** is exposed to the outside. Thus, the first sealing portion **102** is not visible in FIGS. **2**A and **2**B.

The liquid ejection head **100** also includes an electric wiring member **110** for transmitting electric signals from the inkjet recording apparatus to the ejection substrate **101**. An electric signal is input to the ejection substrate **101** from the inkjet recording apparatus through an external signal input terminal **108**. The electric wiring member **110** is fixed to the surface of the channel member **111** so as to surround the recess of the channel member **111** in which the ejection substrate **101** is disposed. The electric wiring member **110** is connected to both ends of the ejection substrate **101** in the Y direction (sides parallel to the X direction), and the connection portions are each sealed from ink and protected by a first connection sealing portion **201** and a second connection sealing portion **202**. As will be described below, the first connection sealing portion **201** is placed to cover the electric wiring and the second connection sealing portion **202**, so the second connection sealing portion **202** is not visible in FIGS. **2**A and **2**B.

Referring to FIG. **1**, the configuration of the liquid ejection head **100** of the first embodiment is described in detail. FIG. **1** is a schematic partial cross-sectional view showing the vicinity of the ejection substrate of the liquid ejection head **100** of the inkjet recording apparatus according to the first embodiment of the present invention. FIG. **1** is a cross-sectional view taken perpendicular to the Y direction (the direction in which a plurality of nozzles is arranged in a nozzle row), and is a cross-sectional view taken along line A-A shown in FIG. **2**A.

The ejection substrate **101** includes a silicon substrate, on which a plurality of thermoelectric conversion elements is formed, and a channel forming member (also referred to as a nozzle plate) formed on the silicon substrate. This channel forming member forms a plurality of liquid channels including liquid chambers surrounding the thermoelectric conversion elements, and a plurality of nozzles **115** communicating with the respective liquid chambers. Also, a common liquid chamber common to the multiple liquid channels is formed in the channel forming member, and ink supply ports each having an elongated rectangular opening extend through the silicon substrate to supply ink to the common liquid chamber. These components are simplified in FIG. **1**. The shape of the opening of the ink supply port is elongated in a direction (Y direction) along line B-B in FIG. **2**A.

In FIG. **1**, the ejection substrate **101** is bonded and fixed to the channel member **111**. The ejection substrate **101** includes nozzle rows each formed by a plurality of nozzles arranged in a first direction (Y direction). The ejection substrate **101** has a configuration in which a plurality of such nozzle rows is arranged in a second direction (X direction). The first direction (Y direction) intersects with the second direction (X direction), and they are perpendicular to each other in the first embodiment. The channel member **111** includes liquid channels **113** for supplying the ink stored in the ink container portion of the liquid ejection head **100** to the ejection substrate **101**. The liquid channels **113** of the channel member **111** are provided so as to correspond to the positions and shapes of the ink supply ports (through holes) of the silicon substrate constituting the ejection substrate **101**. The liquid channels **113** communicate with the nozzles of the ejection substrate **101**. The channel member **111** is made of the same resin material as that of the ink container portion **114** of the liquid ejection head **100** by injection molding using a mold.

The channel member **111** has a recess **116** having a bottom surface to which the ejection substrate **101** is fixed. The electric wiring member **110** is bonded and fixed to the channel member **111** so as to surround the recess **116**.

A space **112** (gap) surrounded by the side surface **117** of the ejection substrate **101** and the side surface **119** and the bottom surface **118** of the recess **116** of the channel member **111** is filled with a first sealing member **120**. In the first embodiment, within the space **112**, a predetermined space located upstream of the ejection substrate **101** in the second direction (X direction) intersecting with the first direction (Y direction) (the space on the right side in FIG. **1**) is filled with the first sealing member **120**. Note that the first sealing member **120** may also be provided in a space within the space **112** that is downstream of the ejection substrate **101** in the second direction (the space on the left side in FIG. **1**).

In the first embodiment, the first sealing member **120** includes a first sealing portion **102** and a second sealing portion **103**. The first sealing portion **102** is positioned in the bottom surface side of the space **112** (the side closer to the bottom surface **118**) and made of a first resin. The second sealing portion **103** is positioned in the ejection surface side of the space **112** (the side closer to the ejection surface **121** of the ejection substrate **101** at which the nozzles **115** open) and made of a second resin. As the first resin and the second resin, thermosetting resins that are relatively easy to handle in the manufacturing process are used. The specific gravity of the first resin constituting the first sealing portion **102** is greater than the specific gravity of the second resin constituting the second sealing portion **103**. As a result, the second sealing portion **103** is positioned above the first sealing portion **102** as viewed in FIG. **1**.

The coefficient of linear expansion of the first resin constituting the first sealing portion **102** is less than the coefficient of linear expansion of the second resin constituting the second sealing portion **103**. The thermosetting resin of the first sealing portion **102** has a small coefficient of linear expansion and thus reduces the tensile stress acting on the ejection substrate **101**, which is in contact with the first sealing portion **102** at the side surface **117**, when the first sealing portion **102** is cured and shrunk after being heated and then cooled. This limits cracking of the ejection substrate **101**.

The surface tension of the second resin constituting the second sealing portion **103** is less than the surface tension of the first resin constituting the first sealing portion **102**. Since the thermosetting resin of the second sealing portion **103** has a small surface tension, the attraction between the particles of the liquid and the wall surface is greater than the attraction in liquid, so that a concave meniscus tends to occur between the liquid surface and the wall surface. Accordingly, the surface **122** of the second sealing portion **103** has a smooth concave shape that is concave toward the bottom surface **118** in a cross section that intersects with the first direction (Y direction). In the first embodiment, the surface **122** of the second sealing portion **103** is formed to connect the upstream end (corner) in the second direction (X direction) of the ejection surface **121** of the ejection substrate **101** and the upstream end in the second direction (X direction) of the recess **116** (end of the channel member **111**). The viscosity of the second resin may be less than the viscosity of the first resin. Also, the thixotropic coefficient of the second resin may be less than the thixotropic coefficient of the first resin.

Furthermore, the filling amount of the first resin constituting the first sealing portion **102** is greater than the filling amount of the second resin constituting the second sealing portion **103**. As a result, the tensile stress acting on the ejection substrate **101** caused by curing shrinkage of the thermosetting resin can be reduced, thereby limiting cracking of the ejection substrate **101**.

The first sealing member **120** is formed as follows. First, the first resin is applied by a dispensing method to the space **112** surrounded by the side surface **117** of the ejection substrate **101** and the bottom surface **118** and the side surface **119** of the recess **116** of the channel member **111**. The shape change of the applied first resin reaches an equilibrium state after a lapse of a certain time, the surface becomes flat, and the first sealing portion **102** is formed. After the surface of the first resin becomes flat, the second sealing portion **103** is formed by applying the second resin onto the first sealing portion **102** by a dispensing method.

FIG. **3** is a diagram showing the area around a connection portion between the ejection substrate and the electric wiring member. FIG. **3** is a cross-sectional view taken perpendicular to the X direction (the direction in which the nozzle rows are arranged), and is a cross-sectional view taken along line B-B shown in FIG. **2**A.

In the first embodiment, the ejection substrate **101** is electrically connected to the electric wiring member **110** for inputting electric signals from the inkjet recording apparatus (outside). Tape automated bonding (TAB) is used for the electric wiring member **110**, and a flying lead **501** of the TAB device is bonded to an electric pad (not shown) on the upper surface of the ejection substrate **101**. The electric pad and the flying lead **501** form a connection portion electrically connecting the electric wiring member **110** and the ejection substrate **101**.

The electric wiring member **110** is attached and fixed to the surface of the channel member **111** so as to surround the recess **116** of the channel member **111**. The electric wiring member **110** and the ejection substrate **101** are connected by flying leads **501** (connection portions) at both ends of the ejection substrate **101** in the first direction (Y direction). That is, the connection portions for electrically connecting the electric wiring member **110** and the ejection substrate **101** are provided at both ends of the ejection substrate **101** in the first direction (Y direction).

The second sealing member **220** covers each connection portion between the electric wiring member **110** and the ejection substrate **101**. The second sealing member **220** includes a first connection sealing portion **201**, which covers the upper portion of the connection portion, and a second connection sealing portion **202**, which covers the lower portion of the connection portion, and protects the connection portion from the ink. The second connection sealing portion **202**, which protects the lower portion of the connection portion, is provided in the space **112** as is the first sealing portion **102**. Within the space **112** surrounded by the bottom surface **118** and the side surface **119** of the recess **116** of the channel member **111** and the side surface **117** of the ejection substrate **101**, the second connection sealing portion **202** is provided in a space located on the upstream side of the ejection substrate **101** in the Y direction. Additionally, within the space **112**, the second connection sealing portion **202** is also provided in a space located downstream of the ejection substrate **101** in the Y direction. The configuration of the second connection sealing portion **202** may be different from that of the first sealing portion **102**.

In the first embodiment, the first sealing portion **102**, the second sealing portion **103**, the first connection sealing portion **201**, and the second connection sealing portion **202**, which seal the space **112** around the ejection substrate **101**, are cured in an oven at 100° C. for 1 hour or more after the application of thermosetting resins. These curing conditions for the thermosetting resins are merely an example. The conditions may be determined according to factors such as ink resistance and adhesive strength, and are not limited to the above example.

FIGS. **4**A to **4**D show the wiping operation of the liquid ejection head **100** described in the first embodiment.

Paper dust **401** generated from the recording paper or the like in the printing process, dust **402** floating in the air, splashes **403** of ink droplets ejected from the ejection ports, and the like adhere to and accumulate on the surfaces of the ejection substrate **101**, the second sealing portion **103**, and the electric wiring member **110**. These deposits (adherents, foreign matter) may affect the ink ejection operation and the ejection performance. For this reason, the inkjet recording apparatus of the first embodiment includes a wiper **300**, which removes the deposits by moving a blade **301** in the second direction (X direction). The blade **301** can be brought into contact with the ejection surface **121** of the ejection substrate **101**, the surface **122** of the second sealing portion **103**, and the surface of the electric wiring member **110**. The blade **301** is made of an elastic material such as rubber. The distal end portion of the blade **301** is brought into contact with the ejection surface **121** and the like and thus deformed, thereby remaining in contact with the ejection surface **121** and the like.

The wiper **300** includes a driving device **302**, which lifts and lowers the blade **301** in the Z directions and also moves the blade **301** in the X directions along a guide member **303** relative to the ejection surface **121** of the ejection substrate **101**.

A wiping operation is performed in the following steps. First, as shown in FIG. **4**A, the driving device **302** moves the blade **301** closer to the channel member **111** on the upstream side in the X direction of the space **112** filled with the first sealing member **120**, and brings the blade **301** into contact with the surface of the electric wiring member **110**. As shown in FIGS. **4**B and **4**C, the driving device **302** then moves the blade **301** in the X direction toward the ejection substrate **101**. Arrow R indicates the moving direction. In the first embodiment, the moving direction R of the blade **301** is parallel to the positive X direction. From a position on the electric wiring member **110** that is located upstream in the second direction (X direction, the moving direction R of the blade **301**) of the space **112** filled with the first sealing member **120**, the blade **301** passes over the surface **122** of the second sealing portion **103** and then passes over the ejection surface **121** of the ejection substrate **101**.

The electric wiring member **110** and the ejection surface **121** of the ejection substrate **101** are connected by the surface **122** of the second sealing portion **103** having a smooth concave shape. Accordingly, during wiping operation, the distal end portion of the blade **301** moves in the X direction while maintaining contact with the surface of the electric wiring member **110**, the surface **122** of the second sealing portion **103**, and the ejection surface **121** of the ejection substrate **101**. As a result, as the blade **301** moves, adherents such as paper dust **401**, dust **402**, and ink droplets **403** can be removed from the surface of the electric wiring member **110**, the surface **122** of the second sealing portion **103**, and the ejection surface **121** of the ejection substrate **101**.

In the first embodiment, since the space **112** (see FIG. **1**) is filled with the first sealing member **120**, paper dust **401**, dust **402**, ink droplets **403** and the like do not accumulate in this space **112**. As such, paper dust, dust, or ink droplets do not accumulate in the space, adhere to the distal end portion of the blade **301**, or remain on the ejection substrate **101** after being moved by the blade **301**.

As described above, the liquid ejection head **100** of the first embodiment limits cracking of the ejection substrate **101** and also reduces the possibility of the ejection performance being affected by foreign matter caused by wiping operation.

Second Embodiment

A second embodiment of the present invention is now described. FIG. **5**A is a schematic partial cross-sectional view showing the vicinity of an ejection substrate of a liquid ejection head **100** of an inkjet recording apparatus according to the second embodiment. FIG. **5**A is a cross-sectional view taken perpendicular to the Y direction (the direction in which a plurality of nozzles is arranged in a nozzle row), and is a cross-sectional view taken along line D-D shown in FIG. **5**B. FIG. **5**B is a plan view showing the vicinity of the ejection substrate of the liquid ejection head **100** of the inkjet recording apparatus according to the second embodiment. FIG. **5**B is a diagram of the liquid ejection head **100** as viewed in the −Z direction. Hereinafter, same reference numerals are given to those components that are the same as those of the first embodiment. The descriptions of such components are omitted as appropriate.

Referring to FIGS. **5**A and **5**B, the liquid ejection head **100** of the second embodiment differs from the first embodiment in the configuration of the first sealing member **120** filling the predetermined space **112**. Specifically, the first sealing member **120** of the second embodiment includes a first sealing portion **102**, a second sealing portion **103**, and a third sealing portion **104**. As in the first embodiment, the first sealing portion **102** is positioned in the bottom surface side of the space **112** (the side closer to the bottom surface **118**) and made of a first resin. The second sealing portion **103** is positioned in the ejection surface side (the side closer to the ejection surface **121**) of the space **112**, in contact with the side surface **117** of the ejection substrate **101**, and made of a second resin. The third sealing portion **104** is positioned in the ejection surface side (the side closer to the ejection surface **121**) of the space **112**, in contact with the side surface **119** of the recess **116**, and made of a third resin. The second resin and the third resin may be the same thermosetting resin or different thermosetting resins. The specific gravity of the first resin constituting the first sealing portion **102** is greater than the specific gravity of the second resin constituting the second sealing portion **103** and the specific gravity of the third resin constituting the third sealing portion **104**. As a result, the second sealing portion **103** and the third sealing portion **104** are located above the first sealing portion **102** as viewed in FIG. **5**A.

The coefficient of linear expansion of the first resin constituting the first sealing portion **102** is less than the coefficient of linear expansion of the second resin constituting the second sealing portion **103** and the coefficient of linear expansion of the third resin constituting the third sealing portion **104**. The thermosetting resin of the first sealing portion **102** has a small coefficient of linear expansion and thus reduces the tensile stress acting on the ejection substrate **101**, which is in contact with the first sealing portion **102** at the side surface **117**, when the first sealing portion **102** is cured and shrunk after being heated and then cooled. This limits cracking of the ejection substrate **101**.

Furthermore, the filling amount of the first resin constituting the first sealing portion **102** is greater than the filling amount of the second resin constituting the second sealing portion **103** and the filling amount of the third resin constituting the third sealing portion **104**. As a result, the tensile stress acting on the ejection substrate **101** caused by curing shrinkage of the thermosetting resin can be reduced, thereby limiting cracking of the ejection substrate **101**.

The surface tension of the second resin constituting the second sealing portion **103** and the surface tension of the third resin constituting the third sealing portion **104** are less than the surface tension of the first resin constituting the first sealing portion **102**. The viscosity of the second resin and the viscosity of the third resin may be less than the viscosity of the first resin. Also, the thixotropic coefficient of the second resin and the thixotropic coefficient of the third resin may be less than the thixotropic coefficient of the first resin.

The second sealing portion **103** has a first inclined surface **123** inclined toward the first sealing portion **102** from the upstream end in the second direction (X direction) of the ejection surface **121** of the ejection substrate **101**. Since the surface tension of the second resin is small, the first inclined surface **123** is a gentle slope of a concave shape toward the first sealing portion **102**.

The third sealing portion **104** has a second inclined surface **124** inclined toward the first sealing portion **102** from the upstream end in the second direction (X direction) of the recess **116** of the channel member **111**. Since the surface tension of the third resin is small, the second inclined surface **24** is a gentle slope of a concave shape toward the first sealing portion **102**.

The upstream end in the second direction (X direction) of the second sealing portion **103** is located downstream in the second direction (X direction) of the downstream end in the second direction (X direction) of the third sealing portion **104**. That is, the upstream end in the second direction (X direction) of the second sealing portion **103** is separated from the downstream end in the second direction (X direction) of the third sealing portion **104**. The second sealing portion **103** and the third sealing portion **104** are not connected. As a result, a part of the first sealing portion **102** is exposed to and thus visible from the outside between the upstream end in the second direction (X direction) of the second sealing portion **103** and the downstream end in the second direction (X direction) of the third sealing portion **104**.

When the channel member **111** has a large coefficient of linear expansion, tensile stress acts on the third sealing portion **104** when the channel member **111** shrinks due to an environmental change or the like. Since the second sealing portion **103** and the third sealing portion **104** are not connected, it is unlikely that the tensile stress of the third sealing portion **104** acts on the ejection substrate **101** through the second sealing portion **103**. As such, the tensile stress generated in the ejection substrate **101** due to the shrinkage of the channel member **111** is reduced, thereby limiting cracking of the ejection substrate **101**.

Also, the shapes of the second sealing portion **103** and the third sealing portion **104** allow the blade **301** of the wiper **300** to pass through the region of the first sealing member **120** not receiving a large resistance. As a result, the wiper **300** can desirably remove the foreign matter adhering to the surface of the first sealing member **120**.

As described above, the liquid ejection head **100** of the second embodiment limits cracking of the ejection substrate **101** in a further reliable manner and also reduces the possibility of the ejection performance being affected by foreign matter caused by wiping operation.

While the present invention has been described with reference to exemplary embodiments, it is to be understood that the invention is not limited to the disclosed exemplary embodiments. The scope of the following claims is to be accorded the broadest interpretation so as to encompass all such modifications and equivalent structures and functions.

This application claims the benefit of Japanese Patent Application No. 2022-126385, filed on Aug. 8, 2022, which is hereby incorporated by reference herein in its entirety.

What is claimed is:

1. A liquid ejection head comprising:

a channel member including a liquid channel;

an ejection substrate disposed on a bottom surface of a recess disposed in the channel member, the ejection substrate including a nozzle row having a plurality of nozzles communicating with the liquid channel, the nozzles being arranged in a first direction; and

a sealing member having a thermosetting resin filled in a predetermined space within a space defined by a side surface of the recess, the bottom surface of the recess, and a side surface of the ejection substrate, and the predetermined space being located upstream of the ejection substrate with respect to a second direction intersecting with the first direction, wherein

the sealing member includes:

a first sealing portion having a first resin located near the bottom surface within the predetermined space; and

a second sealing portion having a second resin located near an ejection surface of the ejection substrate within the predetermined space, the ejection surface being a surface provided with the nozzles,

a coefficient of linear expansion of the first resin is less than a coefficient of linear expansion of the second resin, and

a specific gravity of the first resin is greater than a specific gravity of the second resin.

2. The liquid ejection head according to claim **1**, wherein a surface of the second sealing portion has a concave shape that is concave toward the bottom surface in a cross section intersecting with the first direction.

3. The liquid ejection head according to claim **1**, wherein a surface of the second sealing portion is formed so as to connect an upstream end of the ejection surface with respect to the second direction and an upstream end of the recess with respect to the second direction.

4. The liquid ejection head according to claim **1**, wherein a surface tension of the second resin is less than a surface tension of the first resin.

5. The liquid ejection head according to claim **1**, wherein a filling amount of the first resin constituting the first sealing portion is greater than a filling amount of the second resin constituting the second sealing portion.

6. The liquid ejection head according to claim **1**, wherein the sealing member is a first sealing member,

the liquid ejection head further comprises:

an electric wiring member configured to input an electric signal from an outside to the ejection substrate;

a connection portion electrically connecting the electric wiring member and the ejection substrate; and

a second sealing member covering the connection portion,

the electric wiring member is fixed to a surface of the channel member so as to surround the recess, and

the connection portion connects an end of the ejection substrate in the first direction and the electric wiring member.

7. An inkjet recording apparatus comprising:

the liquid ejection head according to claim **1**; and

a wiper configured to wipe the ejection surface with a blade that can be brought into contact with the ejection surface,

wherein the wiper is configured to move the blade relative to the ejection surface in the second direction such that the blade moves from a position on the channel member upstream of the predetermined space with respect to the second direction, passes over the first sealing portion, and then passes over the ejection surface.

8. A liquid ejection head comprising:

a channel member including a liquid channel;

an ejection substrate disposed on a bottom surface of a recess disposed in the channel member, the ejection substrate including a nozzle row having a plurality of nozzles communicating with the liquid channel, the nozzles being arranged in a first direction; and

a sealing member having a thermosetting resin filled in a predetermined space within a space defined by a side surface of the recess, the bottom surface of the recess, and a side surface of the ejection substrate, the predetermined space being located upstream of the ejection substrate with respect to a second direction intersecting with the first direction, wherein

the sealing member includes:

a first sealing portion having a first resin located near the bottom surface within the predetermined space;

a second sealing portion having a second resin located near an ejection surface of the ejection substrate within the predetermined space, the ejection surface being a surface provided with the nozzles, and the second sealing portion being in contact with the side surface of the ejection substrate; and

a third sealing portion having a third resin located near the ejection surface of the ejection substrate within the predetermined space, the third sealing portion being in contact with the side surface of the recess,

a coefficient of linear expansion of the first resin is less than a coefficient of linear expansion of the second resin and a coefficient of linear expansion of the third resin, and

a specific gravity of the first resin is greater than a specific gravity of the second resin and a specific gravity of the third resin.

9. The liquid ejection head according to claim **8**, wherein

the second sealing portion includes a first inclined surface inclined toward the first sealing portion from an upstream end of the ejection surface with respect to the second direction, and

the third sealing portion includes a second inclined surface inclined toward the first sealing portion from an upstream end of the recess with respect to the second direction.

10. The liquid ejection head according to claim **8**, wherein

an upstream end of the second sealing portion with respect to the second direction is located downstream in the second direction of a downstream end of the third sealing portion with respect to the second direction, and

a part of the first sealing portion is exposed between the upstream end of the second sealing portion and the downstream end of the third sealing portion.

11. The liquid ejection head according to claim **8**, wherein surface tensions of the second resin and the third resin are less than a surface tension of the first resin.

12. The liquid ejection head according to claim **8**, wherein a filling amount of the first resin constituting the first sealing portion is greater than a filling amount of the second resin constituting the second sealing portion and a filling amount of the third resin constituting the third sealing portion.

13. The liquid ejection head according to claim **8**, wherein the sealing member is a first sealing member,

the liquid ejection head further comprises:

an electric wiring member configured to input an electric signal from an outside to the ejection substrate;

a connection portion electrically connecting the electric wiring member and the ejection substrate; and

a second sealing member covering the connection portion,

the electric wiring member is fixed to a surface of the channel member so as to surround the recess, and

the connection portion connects an end of the ejection substrate in the first direction and the electric wiring member.

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